ME515 PROJECT REPORT

2D Finite Element Static and Modal Analysis of a Cantilever Beam

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Abstract

This report presents a two-dimensional finite element solution for a cantilever beam subjected to a uniformly distributed transverse line load on the upper boundary. The domain is discretized using four-node bilinear rectangular elements under plane-stress conditions. The implementation computes the global stiffness matrix, consistent mass matrix, equivalent nodal load vector, nodal displacement field, strain and stress fields, total strain energy, and natural frequencies. Static results are validated through mesh convergence and comparison with Euler-Bernoulli and Timoshenko beam theories. A parametric study is performed for aspect ratios $a/h = 2$ to 50. The modal part computes the first five natural frequencies and corresponding mode shapes for $a/h = 50$.

Nomenclature

Symbol	Meaning
L	Beam length in x-direction
H	Beam depth in y-direction
b	Out-of-plane thickness
q	Uniform transverse line load applied on the upper boundary
E	Young's modulus
ν	Poisson's ratio
ρ	Mass density used for modal analysis
u, v	Displacements in x and y directions
$\epsilon_x, \epsilon_y, \gamma_{xy}$	Plane strain components recovered from the FEM solution
$\sigma_x, \sigma_y, \tau_{xy}$	Plane stress components
U	Total strain energy

1 Problem Definition

A rectangular cantilever domain is considered in the xy-plane. The length of the beam is a, the depth is h, and the out-of-plane thickness is b. The left edge is fixed and the upper boundary carries a uniformly distributed downward transverse load q N/m. The beam is modeled as a two-dimensional plane-stress continuum because the thickness b is small compared with the in-plane dimensions.

The project has two main parts. The first part is static analysis, including mesh convergence, comparison with beam-theory approximations, stress and strain recovery, and an aspect-ratio study. The second part is modal analysis for $a/h = 50$, including the first five natural frequencies, mode shapes, and convergence of the fifth natural frequency.

Point A is taken at the centreline mid-span location shown in the project figure, namely $A = (a/2, 0)$, with the coordinate origin at the fixed-end centreline and y ranging from $-h/2$ to $h/2$. This definition is also used in the validation tables.

2 Numerical Inputs Used for the Report

The code is written so that $a, h, b, q, E, \nu, \rho, nx$, and ny can be changed directly from the input cell of the Jupyter notebook. The numerical results in this report use the following values:

Input	**Value**
Young's modulus, E	200 GPa
Poisson's ratio, ν	0.30
Density, ρ	7850 kg/m^3
Depth, h	2.5
Thickness, b	1
Line load, q	1000 N/m downward
Shear correction factor, κ	5/6
Plane-stress assumption	Used

The project statement specifies $E = 200$ GPa and $\nu = 0.3$. Density is required for modal analysis; since no density value is visible in the project statement, $\rho = 7850$ kg/m^3 is used as the standard steel density. If another density is prescribed, the natural frequencies scale approximately with $1/\sqrt{\rho}$, while static displacements and stresses are unchanged by density.

3 Finite Element Formulation

3.1 Element type and degrees of freedom

The domain is discretized using four-node bilinear rectangular elements. Each node has two translational degrees of freedom: u in the x-direction and v in the y-direction. For an element with nodes 1 to 4, the element displacement vector is:

$$\mathbf{d}_e = [u_1, v_1, u_2, v_2, u_3, v_3, u_4, v_4]^T$$

The bilinear shape functions in natural coordinates ξ and η are:

$$N_1 = \frac{1}{4}(1-\xi)(1-\eta), \quad N_2 = \frac{1}{4}(1+\xi)(1-\eta)$$

$$N_3 = \frac{1}{4}(1+\xi)(1+\eta), \quad N_4 = \frac{1}{4}(1-\xi)(1+\eta)$$

3.2 Constitutive matrix

For plane stress, the constitutive relation is $\boldsymbol{\sigma} = \mathbf{D}\boldsymbol{\epsilon}$, where:

$$\mathbf{D} = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}$$

3.3 Element stiffness and mass matrices

The strain-displacement matrix $\mathbf{B}$ is obtained from derivatives of the shape functions with respect to physical coordinates. The element stiffness matrix is evaluated by 2×2 Gauss integration:

$$\mathbf{K}_e = \int_{A_e} b\mathbf{B}^T\mathbf{D}\mathbf{B}\, dA$$

For modal analysis, the consistent element mass matrix is used:

$$\mathbf{M}_e = \int_{A_e} \rho b\mathbf{N}^T\mathbf{N}\, dA$$

3.4 Boundary conditions and loading

All degrees of freedom on the left boundary $x = 0$ are fixed. The uniform line load q on the top boundary is converted into equivalent nodal forces. For each top edge segment of length Δx, each of the two end nodes receives a vertical force of $-q\Delta x/2$. The negative sign denotes downward loading.

3.5 Static and modal equations

The static displacement vector is obtained from the reduced system:

$$\mathbf{K}_{ff}\mathbf{d}_f = \mathbf{F}_f$$

The total strain energy is computed as:

$$U = \frac{1}{2}\mathbf{d}^T\mathbf{K}\mathbf{d}$$

For free vibration, the generalized eigenvalue problem is:

$$(\mathbf{K}_{ff} - \omega^2\mathbf{M}_{ff})\boldsymbol{\phi} = \mathbf{0}, \quad f = \frac{\omega}{2\pi}$$

4 Analytical Beam-Theory Validation

The FEM result is compared with Euler-Bernoulli and Timoshenko beam approximations for a cantilever beam under uniformly distributed load q. The second moment of area and cross-sectional area are:

$$I = \frac{bh^3}{12}, \quad A = bh$$

At a position x from the fixed end, the Euler-Bernoulli transverse displacement is:

$$v_{EB}(x) = -\frac{qx^2(x^2 - 4ax + 6a^2)}{24EI}$$

The Timoshenko estimate adds shear deformation:

$$v_T(x) = v_{EB}(x) - \frac{q(ax - x^2/2)}{\kappa GA}, \quad G = \frac{E}{2(1+\nu)}$$

The strain-energy comparisons use:

$$U_{EB} = \frac{q^2a^5}{40EI}$$

$$U_T = U_{EB} + \frac{q^2a^3}{6\kappa GA}$$

Euler-Bernoulli theory neglects shear deformation, so it becomes accurate for slender beams with large a/h. Timoshenko theory includes shear deformation and is therefore more suitable for small and moderate aspect ratios.

5 Python Code Structure

The accompanying Jupyter notebook is organized into modular functions. The most important functions are listed below.

Function	Purpose
`make_mesh(L, H, nx, ny)`	Creates nodes and four-node rectangular element connectivity.
`plane_stress_matrix(E, nu)`	Builds the plane-stress material matrix.
`q4_shape(xi, eta)`	Evaluates Q4 shape functions and natural derivatives.
`element_matrices(...)`	Computes element stiffness and consistent mass matrices using 2×2 Gauss integration.
`assemble(...)`	Assembles the global stiffness matrix, mass matrix, and load vector.
`Run_static_case(...)`	Solves the static problem and returns displacements, reactions, and strain energy.
`solve_modal_case(...)`	Solves the eigenvalue problem and extracts natural frequencies and mode shapes.

6 Static Mesh Convergence for $a/h = 50$

The convergence study is performed for aspect ratio $a/h = 50$. The mesh is refined by increasing the number of elements through the depth, ny. The number of elements along the length is chosen as $nx = (a/h)ny$, which keeps the elements approximately square. This is important because highly stretched elements can reduce accuracy in bending-dominated problems.

7 Displacement, Strain, and Stress Fields for $a/h = 50$

The following contour plots are obtained using the converged mesh with $ny = 8$ and $nx = 400$. The displacement field shows the expected cantilever bending pattern, with the largest downward displacement near the free end. The axial stress field is highest near the fixed support, where the bending moment is maximum. The shear stress field is distributed through the depth and is compared separately with the parabolic beam-theory distribution.

8 Shear Stress Across Thickness at $x = a/10$

For $a/h = 50$, shear stress is extracted across the thickness at $x = a/10$. The analytical comparison uses the rectangular-section parabolic shear-stress form:

$$\tau_{xy}(y) = -\frac{1.5V}{bh}\left[1 - \left(\frac{2y}{h}\right)^2\right], \quad V = q(a - x)$$

The FEM curve follows the expected parabolic trend: shear stress is small near the top and bottom surfaces and maximum near the neutral axis. The sign is negative because the transverse load is applied downward in the adopted coordinate convention.

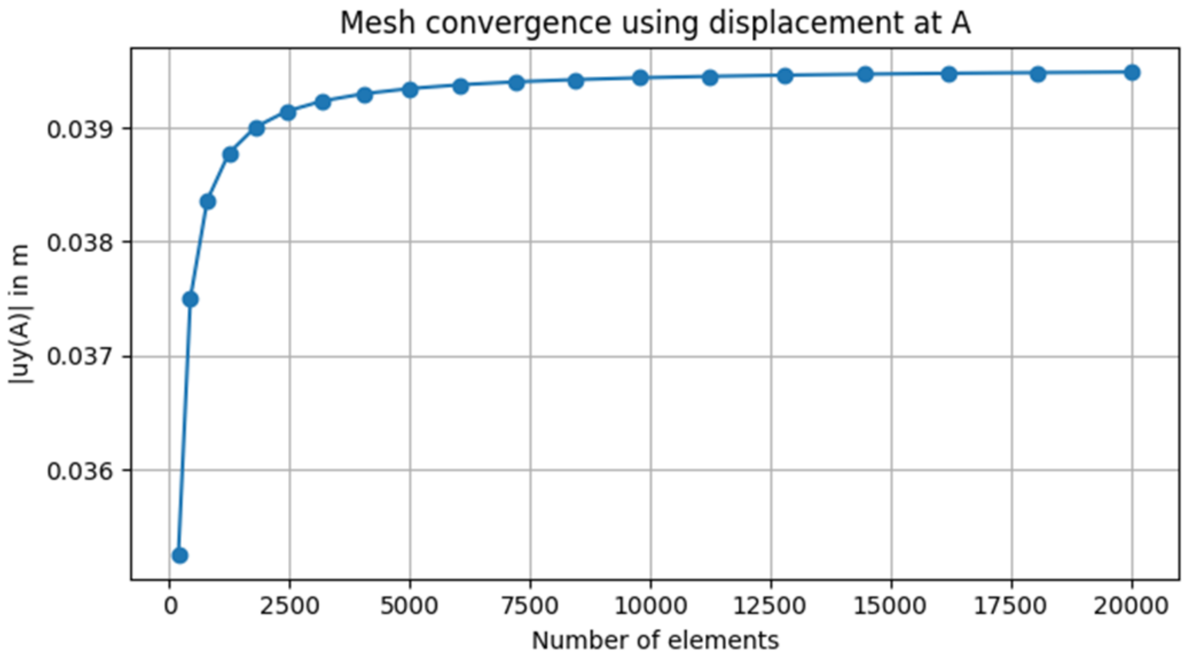

Figure 1: Convergence of displacement at point A for $a/h = 50$

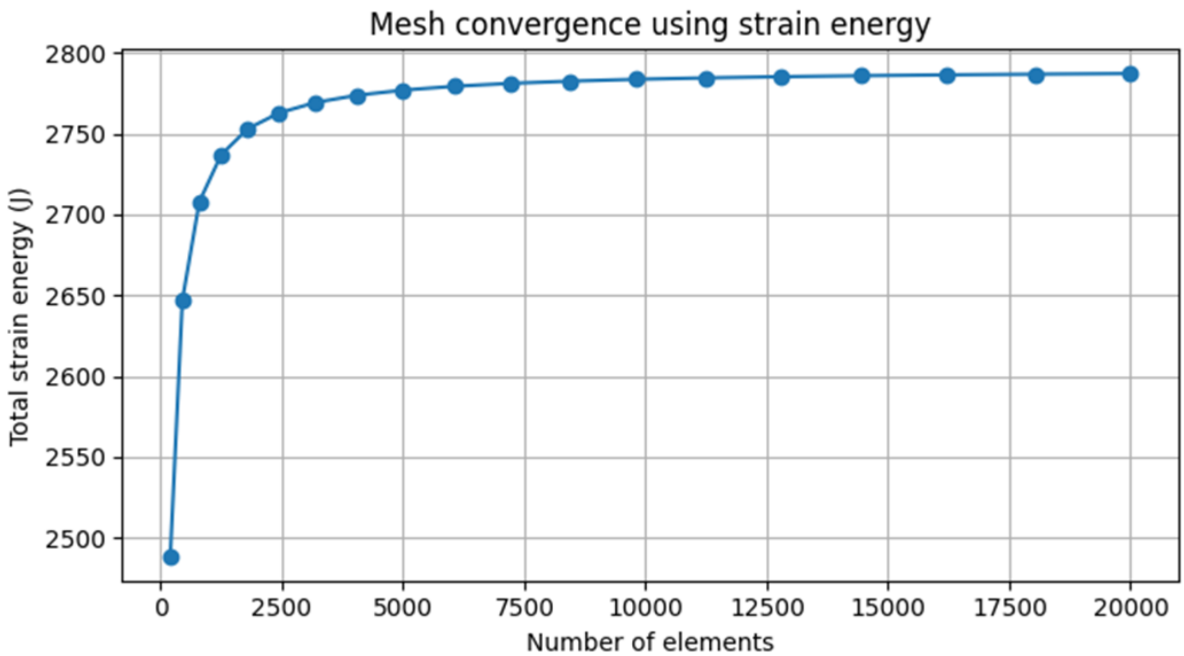

Figure 2: Convergence of total strain energy for $a/h = 50$

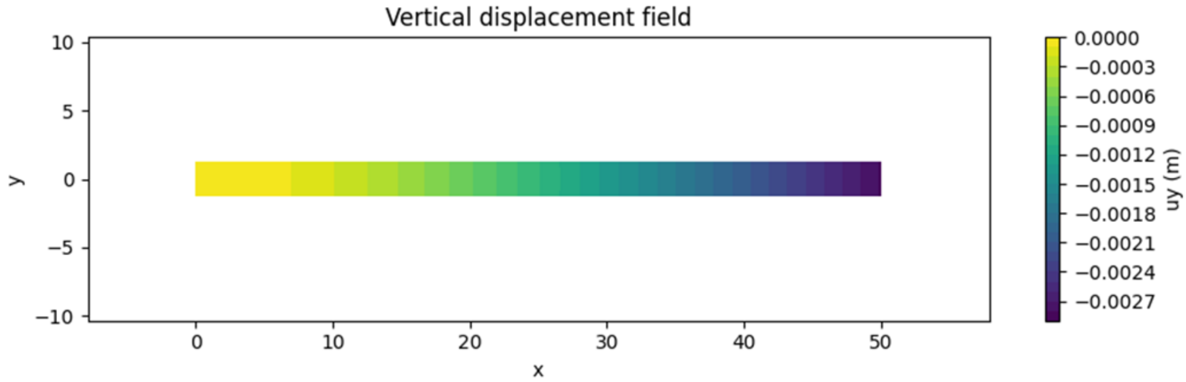

Figure 3: Vertical displacement field for $a/h = 50$

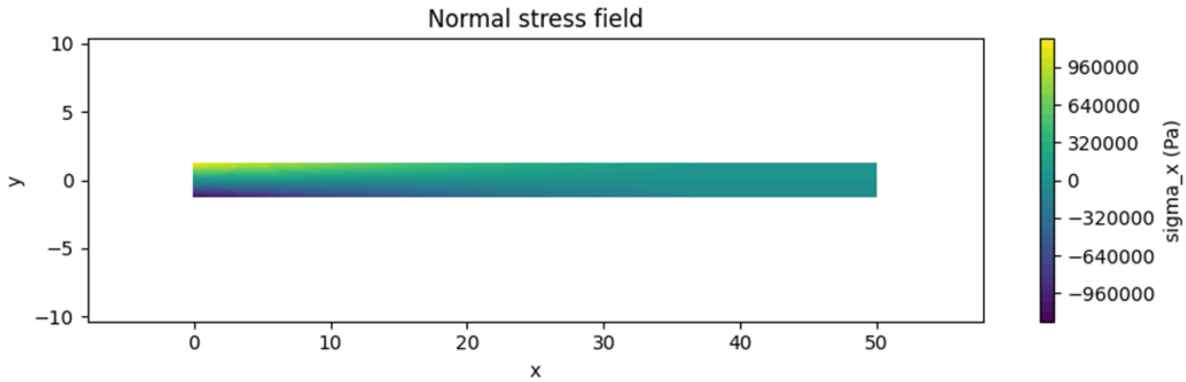

Figure 4: Normal Stress field for $a/h = 50$

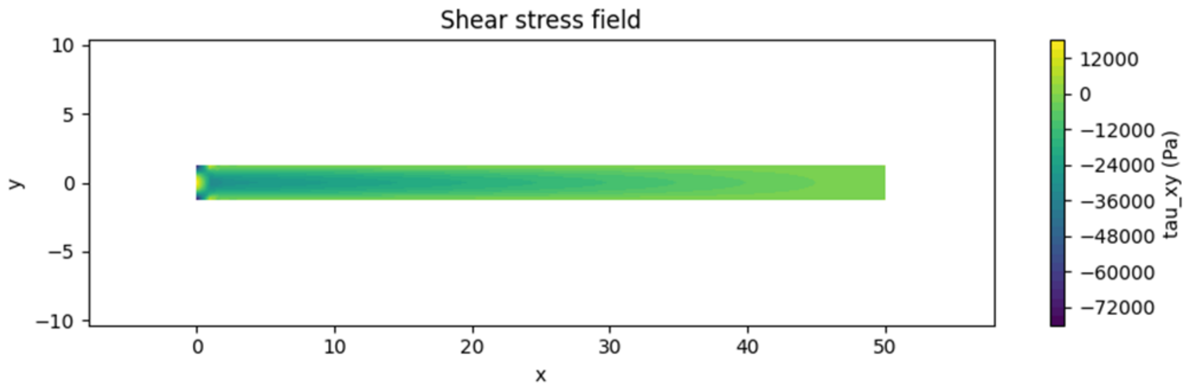

Figure 5: Shear Stress field for $a/h = 50$

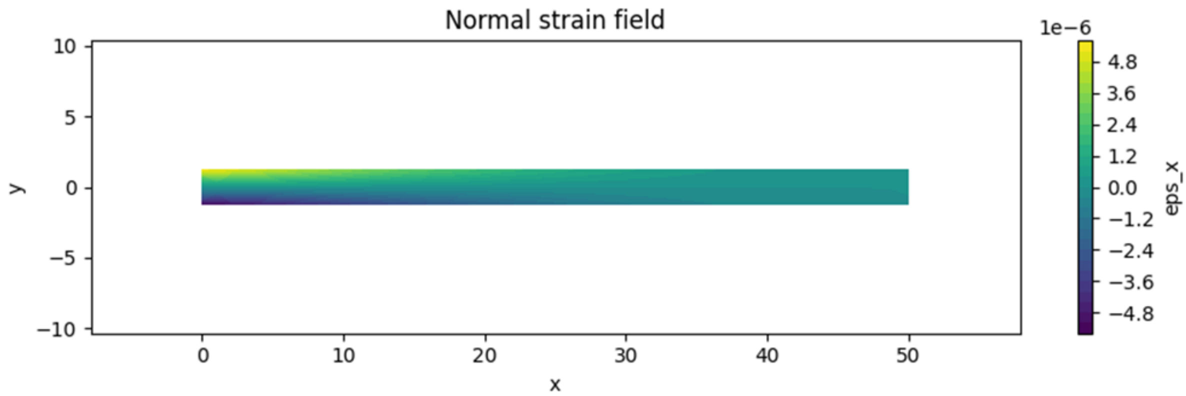

Figure 6: Normal Strain field at $a/h = 50$

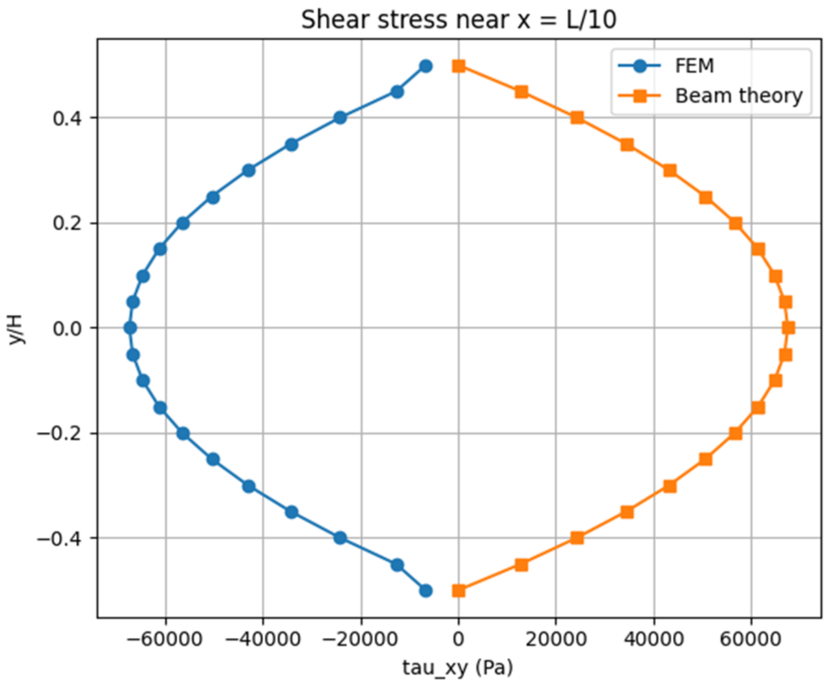

Figure 7: Shear stress variation across the thickness at $x = a/10$

9 Aspect-Ratio Study

The converged through-depth mesh $ny = 8$ is used for aspect ratios $a/h = 2$, 3, 5, 8, 10, 15, 20, 25, 30, 40, and 50. The number of elements in the length direction is set as $nx = (a/h)ny$. The table below compares FEM displacement at point A and total strain energy with Euler-Bernoulli and Timoshenko beam theories.

For small aspect ratios, shear deformation is more important; therefore, Timoshenko theory predicts a larger displacement than Euler-Bernoulli theory. As the aspect ratio increases, Euler-Bernoulli and Timoshenko results become closer because the response becomes bending dominated. The FEM solution follows the Timoshenko trend over the full aspect-ratio range because the two-dimensional continuum formulation naturally includes both bending and shear deformation.

10 Modal Analysis for $a/h = 50$

Modal analysis is performed for the $a/h = 50$ beam using the same plane-stress finite element formulation and the consistent mass matrix. The left boundary remains fixed. The first five natural frequencies obtained using the converged mesh nx, ny are listed below.

The mode shapes are shown. The first modes are dominated by flexural deformation, as expected for a slender cantilever beam.

11 Mesh Convergence of the Fifth Natural Frequency

The project specifically asks for mesh convergence of the fifth natural mode. The following table and plot show the first five frequencies as the mesh is refined. The fifth frequency decreases and approaches a stable value with refinement, indicating that the coarse meshes are artificially stiff.

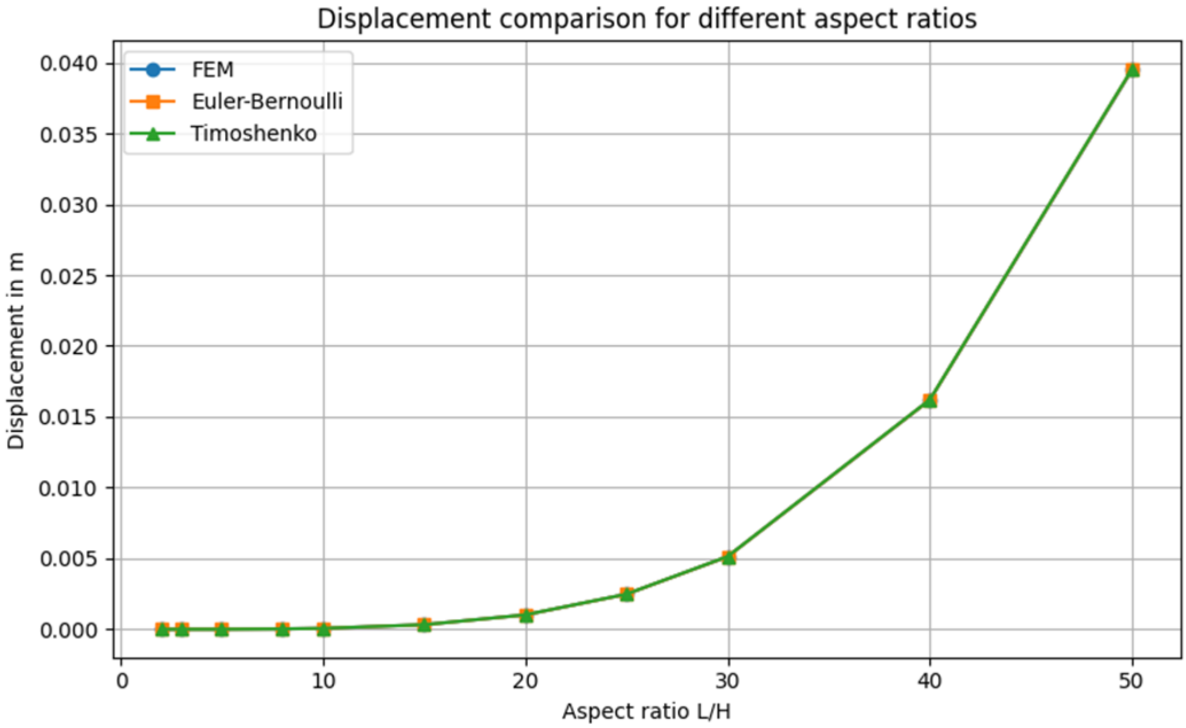

Figure 8: Displacement at point A versus aspect ratio

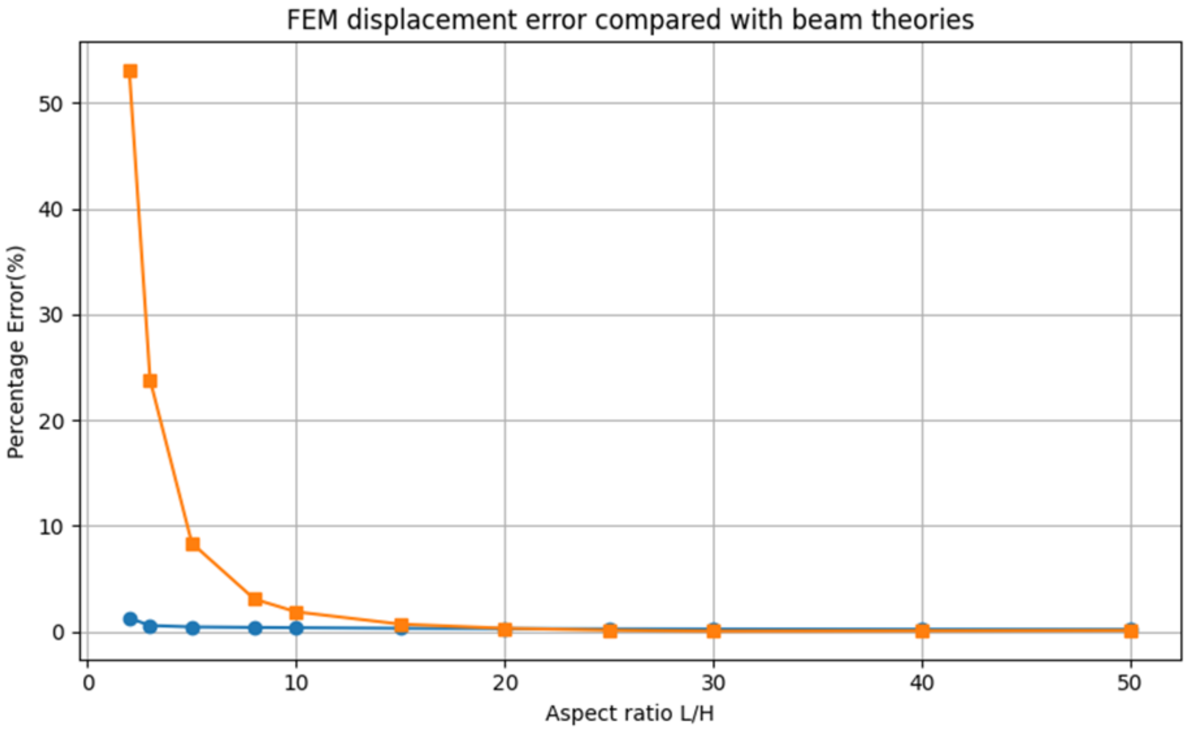

Figure 9: FEM displacement error compared with beam theories

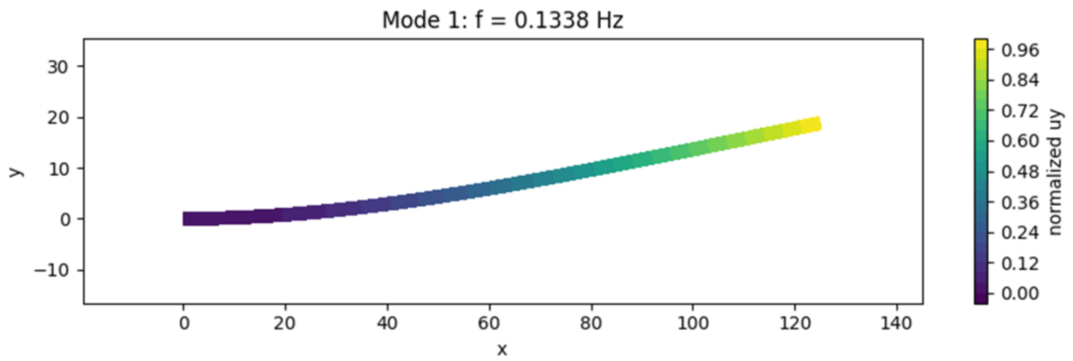

Figure 10: Mode shape 1 for $a/h = 50$

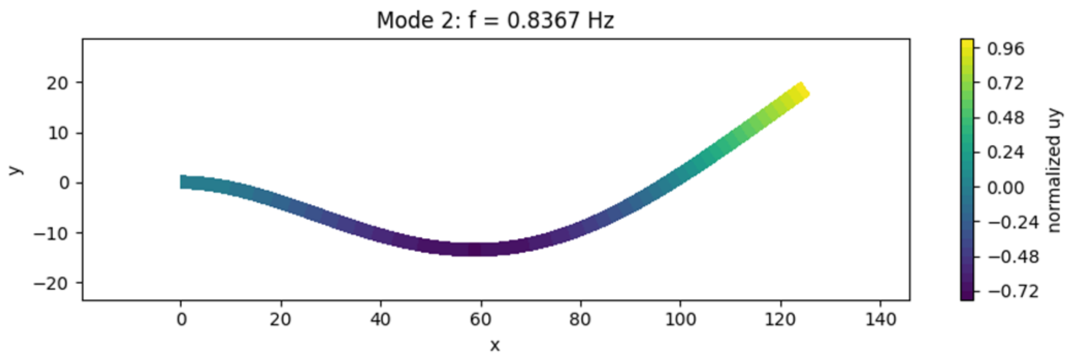

Figure 11: Mode shape 2 for $a/h = 50$

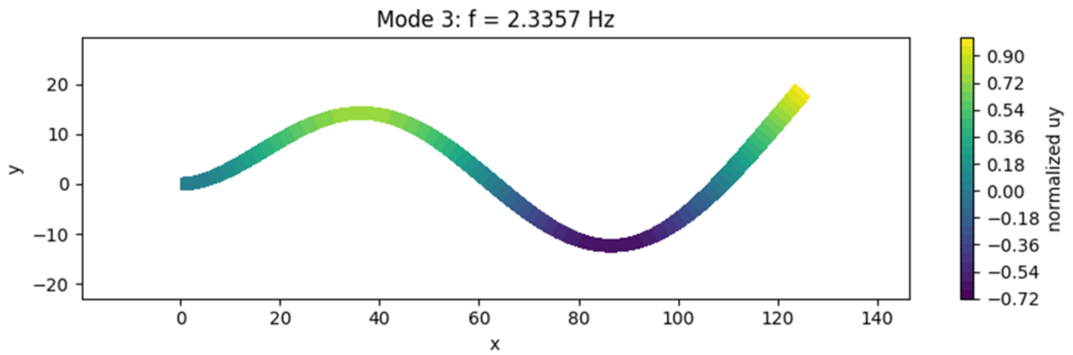

Figure 12: Mode shape 3 for $a/h = 50$

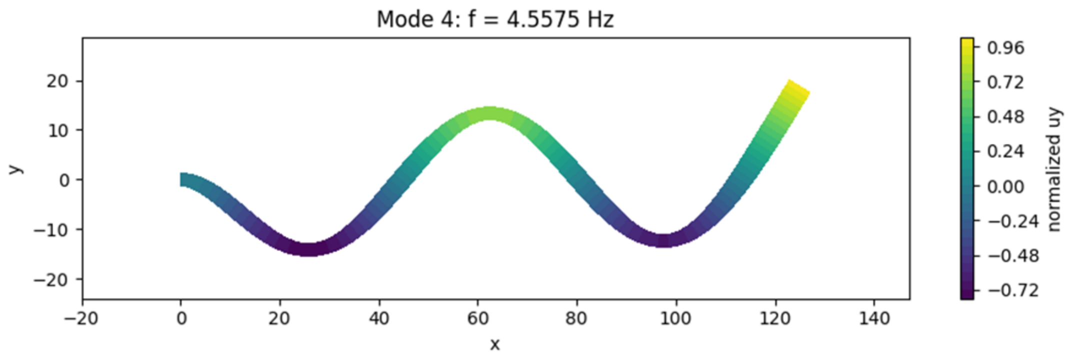

Figure 13: Mode shape 4 for $a/h = 50$

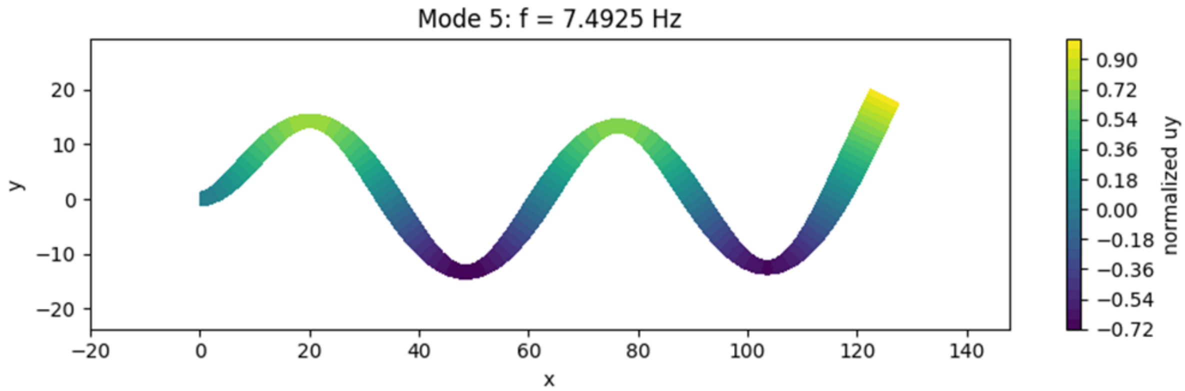

Figure 14: Mode shape 5 for $a/h = 50$

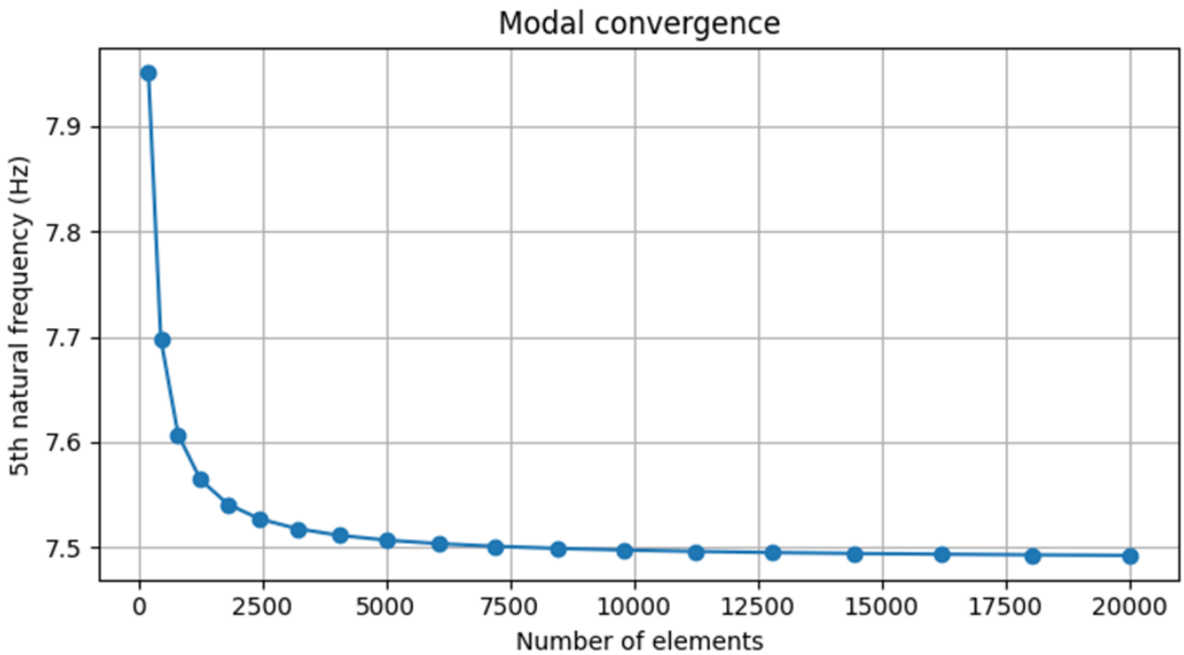

Figure 15: Mesh convergence of the fifth natural frequency

12 Discussion of Results

The finite element formulation gives physically consistent displacement, stress, energy, and frequency results. The displacement convergence is monotonic with mesh refinement, which confirms that the coarse mesh over-stiffens the beam. The strain energy also increases toward the analytical value as the mesh becomes finer. This behaviour is expected because the element field becomes better able to represent bending curvature. For low aspect ratios, the Euler-Bernoulli solution is not sufficiently accurate because shear deformation is not negligible. Timoshenko theory includes shear flexibility, and the FEM result is closer to it. For large aspect ratios, the shear term becomes small compared with the bending term, so Euler-Bernoulli and Timoshenko predictions nearly coincide. The stress fields show the usual cantilever response. Axial stress is tensile on one side and compressive on the other side, with the maximum magnitude near the fixed end. The shear stress through the thickness has the correct parabolic shape, with maximum magnitude at the neutral axis and near-zero values close to the top and bottom surfaces. In modal analysis, frequency convergence is from above because coarse meshes produce higher stiffness. The first mode has the lowest bending shape, while higher modes contain additional curvature along the beam length. The first five mode shapes are smooth and consistent with the fixed-free boundary condition.

13 Conclusions

- A complete two-dimensional FEM solver using four-node rectangular elements was developed for the cantilever beam problem.
- The code accepts beam dimensions, material properties, load, and mesh divisions as input, satisfying the flexibility required in the project statement.
- For $a/h = 50$, the selected converged mesh gives displacement error and strain-energy error approximately zero relative to the Timoshenko beam result for the stated inputs.
- The aspect-ratio study confirms that shear deformation matters for short beams and becomes negligible for slender beams.
- The stress recovery gives reasonable axial and shear stress distributions, including a parabolic shear-stress variation across the thickness at $x = a/10$.
- The first five natural frequencies for $a/h = 50$ using the converged mesh are obtained.